

Reproducibility Report: Maglio & Polman (2014), Study 1

Minimal version

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Goals

1. Load a published dataset with `here()` and `read_excel()`
2. Reproduce the ANOVA and effect sizes reported in the paper
3. Say clearly what did and did not reproduce, and why

This is the **minimal** version: the structure and the questions are here, and you write the code. If you get stuck, `reproducibility-report-bumper-rails.qmd` has the same steps with most of the code filled in. It is fine to peek at it for one step and then come back.

error: `true` is set at the top of this file, so the document still renders while some chunks are unfinished. The chunks that still fail show their error message in red.

Before you start, read Study 1 in the paper: [original_paper/maglio-polman-2014.pdf](#) (two pages).

The study

Maglio, S. J., & Polman, E. (2014). Spatial orientation shrinks and expands psychological distance. *Psychological Science*, 25(7), 1345–1352. <https://doi.org/10.1177/0956797614530571>

Methods summary. Researchers recruited 202 volunteers at the Bay Street subway station in Toronto. Half were waiting on the westbound platform and half on the eastbound platform. Each participant was randomly assigned to rate how far away one of four stations felt (1 = very close, 7 = very far): Spadina (two stops west), St. George (one stop west), Bloor-Yonge (one stop east), or Sherbourne (two stops east). The authors ran a 2×4 ANOVA on the distance ratings, then compared eastbound and westbound travelers separately at each station.

Target outcomes. This is the result you are trying to reproduce, quoted from the Study 1 results section:

We carried out a 2 (orientation: toward, away from) $\times$ 4 (station: Spadina, St. George, Bloor-Yonge, Sherbourne) analysis of variance (ANOVA) on closeness ratings, which revealed no main effect of orientation, $F < 1$, and a main effect of station, $F(3, 194) = 24.10$, $p < .001$, $p^2 = .27$. This main effect was qualified by the predicted interaction between orientation and station, $F(3, 194) = 16.28$, $p < .001$, $p^2 = .20$. We decomposed this interaction by the subjective-distance ratings between participants traveling east and west for each of the four subway stations. Westbound participants rated the stations to the west of Bay Street as closer than did eastbound participants; this effect was obtained for both the station one stop to the west (St. George, $p < .001$, $p^2 = .28$) and the station two stops to the west (Spadina, $p = .001$, $p^2 = .20$). The opposite pattern held true for stations to the east of Bay Street. Eastbound participants rated the stations to the east of Bay Street as closer than did westbound participants; this effect was obtained for both the station one stop to the east (Bloor-Yonge, $p = .053$, $p^2 = .08$) and the station two stops to the east (Sherbourne, $p < .001$, $p^2 = .24$). Figure 1 summarizes these results.

p^2 is **partial eta squared**, an effect size. You will compute it below.

Step 1: Load packages

`readxl` (reads Excel files) and `broom` (turns model output into tibbles) are installed along with the tidyverse, but `library(tidyverse)` does not load them, so we load them by name.

```
install.packages(c("tidyverse", "here"))
```

```
library(tidyverse)
library(readxl)
library(broom)
library(here)
```

Step 2: Load the data

The authors posted one Excel file per study. We only need Study 1. The columns are described in `docs/codebook.md`.

```
d <- read_excel(here("data", "S1_Subway.xlsx"))

glimpse(d)
```

Rows: 202

Columns: 4

```
$ DIRECTION <chr> "EAST", "EAST", "EAST", "EAST", "EAST", "EAST", "EAST", "EA~
$ DISTANCE  <dbl> 1, 2, 2, 3, 1, 1, 2, 1, 3, 2, 1, 1, 2, 1, 3, 1, 2, 1, 1, 1,~
$ STN_NUMBER <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3,~
$ STN_NAME  <chr> "B-Y", "B-Y", "B-Y", "B-Y", "B-Y", "B-Y", "B-Y", "B-
Y", "B--
```

Step 3: Tidy the data

The data are already tidy: one row per participant, one column per variable. But R has read `DIRECTION` and `STN_NAME` in as text. For an ANOVA they need to be **factors**, and we set the order of the station levels to run west to east, so tables and plots come out in map order.

Use `mutate()` and `factor()` to convert both columns. For `STN_NAME`, give `factor()` a `levels` argument listing the stations from west to east.

```
d.tidy = d %>%
  mutate(DIRECTION.f = factor(DIRECTION),
         STN_NAME.f = factor(STN_NAME,
                             levels = c("SPAD", "STG",
                                           "B-Y", "SHER")))
```

Step 4: Reproduce the analysis

How many people are in each cell?

Count the participants in each direction $\times$ station combination. Does it match the paper (202 total, 101 per platform)? Is every cell the same size?

```
d.tidy %>%  
  count(DIRECTION.f, STN_NAME.f)
```

```
# A tibble: 8 x 3  
  DIRECTION.f STN_NAME.f      n  
    <fct>      <fct>    <int>  
1 EAST      SPAD        26  
2 EAST      STG         26  
3 EAST      B-Y         23  
4 EAST      SHER        26  
5 WEST      SPAD        25  
6 WEST      STG         25  
7 WEST      B-Y         26  
8 WEST      SHER        25
```

The 2×4 ANOVA

Fit a two-way ANOVA with `aov()`: distance predicted by direction, station, and their interaction. Remember that `y ~ A * B` is shorthand for `y ~ A + B + A:B`.

```
main.anova = aov(DISTANCE ~ DIRECTION.f * STN_NAME.f,  
                 data = d.tidy)  
  
summary(main.anova)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
DIRECTION.f	1	0.71	0.713	0.664	0.416
STN_NAME.f	3	75.16	25.053	23.349	6.01e-13 ***
DIRECTION.f:STN_NAME.f	3	52.41	17.471	16.283	1.77e-09 ***
Residuals	194	208.15	1.073		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The numbers in parentheses in “F(3, 194)” are the degrees of freedom: 3 for the effect, 194 for the residuals. Check that yours match.

The teaching team gets:

- Main effect of direction: $F(1, 194) = 0.66$
- Main effect of station: $F(3, 194) = 23.35$
- Interaction between direction and station: $F(3, 194) = 16.28$

Two of those match the paper. One is close but not the same. Keep going; the bonus question at the end asks you to work out why.

Partial eta squared

Partial eta squared is the share of variance an effect accounts for, once the other effects are set aside:

$$\eta_p^2 = \frac{SS_{\text{effect}}}{SS_{\text{effect}} + SS_{\text{residual}}}$$

Compute it for each effect. Hint: `tidy()` from `broom` turns the ANOVA table into a tibble, with the sums of squares in a column called `sumsq`.

```
main.anova.table = tidy(main.anova)
main.anova.table
```

```
# A tibble: 4 x 6
  term                df  sumsq meansq statistic  p.value
  <chr>              <dbl> <dbl> <dbl>    <dbl>    <dbl>
1 DIRECTION.f         1  0.713  0.713    0.664 4.16e- 1
2 STN_NAME.f          3 75.2   25.1   23.3   6.01e-13
3 DIRECTION.f:STN_NAME.f 3 52.4   17.5   16.3   1.77e- 9
4 Residuals         194 208.    1.07    NA     NA
```

```
ss_residual = main.anova.table %>%
  filter(term == "Residuals") %>%
  pull(sumsq)

main.anova.table %>%
  filter(term != "Residuals") %>%
  mutate(partial_eta_sq = (sumsq)/(sumsq + ss_residual)) %>%
  select(term, df, statistic, p.value, partial_eta_sq)
```

```
# A tibble: 3 x 5
  term                df statistic  p.value partial_eta_sq
  <chr>              <dbl>    <dbl>    <dbl>         <dbl>
1 DIRECTION.f         1     0.664 4.16e- 1         0.00341
2 STN_NAME.f          3    23.3  6.01e-13         0.265
3 DIRECTION.f:STN_NAME.f 3    16.3  1.77e- 9         0.201
```

Do your values round to the paper's .27 (station) and .20 (interaction)?

Station-by-station comparisons

Now compare eastbound and westbound travelers at each station separately. For each station: keep only that station's rows, compute the mean rating in each direction, run a one-way ANOVA of distance on direction, and compute p^2 .

St. George

```
##only STG rows
d.stg = d.tidy %>%
  filter(STN_NAME.f == "STG")

##mean distance rating for each direction
d.stg %>%
  group_by(DIRECTION.f) %>%
  summarise(mean_rating = mean(DISTANCE),
            n = n())
```

```
# A tibble: 2 x 3
  DIRECTION.f mean_rating    n
  <fct>        <dbl> <int>
1 EAST         2.77    26
2 WEST         1.64    25
```

```
##running ANOVA
stg.anova = aov(DISTANCE ~ DIRECTION.f, data = d.stg)

summary(stg.anova)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
DIRECTION.f	1	16.25	16.252	18.79	7.23e-05 ***
Residuals	49	42.38	0.865		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
##partial eta squared
stg.table = tidy(stg.anova)
stg.table
```

```
# A tibble: 2 x 6
  term          df sumsq meansq statistic    p.value
<chr>      <dbl> <dbl>  <dbl>    <dbl>    <dbl>
1 DIRECTION.f      1  16.3  16.3      18.8 0.0000723
2 Residuals      49  42.4   0.865     NA    NA
```

```
ss_residual_stg = stg.table %>%
  filter(term == "Residuals") %>%
  pull(sumsq)

stg.table %>%
  filter(term != "Residuals") %>%
  mutate(partial_eta_sq = (sumsq)/(sumsq + ss_residual_stg)) %>%
  select(term, df, statistic, p.value, partial_eta_sq)
```

```
# A tibble: 1 x 5
  term          df statistic    p.value partial_eta_sq
<chr>      <dbl>    <dbl>    <dbl>    <dbl>
1 DIRECTION.f      1      18.8 0.0000723      0.277
```

Spadina

```
##only SPAD rows
d.spad = d.tidy %>%
  filter(STN_NAME.f == "SPAD")

##mean distance rating for each direction
d.spad %>%
  group_by(DIRECTION.f) %>%
  summarise(mean_rating = mean(DISTANCE),
            n = n())
```

```
# A tibble: 2 x 3
  DIRECTION.f mean_rating    n
  <fct>         <dbl> <int>
1 EAST          3.65     26
2 WEST          2.64     25
```

```
##running ANOVA
spad.anova = aov(DISTANCE ~ DIRECTION.f, data = d.spad)

summary(spad.anova)
```

```
              Df Sum Sq Mean Sq F value    Pr(>F)
DIRECTION.f   1   13.10   13.100    11.97 0.00113 **
Residuals    49   53.64    1.095
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##partial eta squared
spad.table = tidy(spad.anova)
spad.table
```

```
# A tibble: 2 x 6
  term          df sumsq meansq statistic p.value
  <chr>        <dbl> <dbl>  <dbl>    <dbl>    <dbl>
1 DIRECTION.f     1  13.1   13.1     12.0 0.00113
2 Residuals     49  53.6    1.09      NA      NA
```

```
ss_residual_spad = spad.table %>%
  filter(term == "Residuals") %>%
  pull(sumsq)

spad.table %>%
  filter(term != "Residuals") %>%
  mutate(partial_eta_sq = (sumsq)/(sumsq + ss_residual_spad)) %>%
  select(term, df, statistic, p.value, partial_eta_sq)
```

```
# A tibble: 1 x 5
  term          df statistic p.value partial_eta_sq
  <chr>        <dbl>    <dbl>    <dbl>         <dbl>
1 DIRECTION.f     1     12.0 0.00113         0.196
```

Bloor-Yonge

```
##only B-Y rows
d.by = d.tidy %>%
  filter(STN_NAME.f == "B-Y")

##mean distance rating for each direction
d.by %>%
  group_by(DIRECTION.f) %>%
  summarise(mean_rating = mean(DISTANCE),
            n = n())
```

```
# A tibble: 2 x 3
  DIRECTION.f mean_rating     n
  <fct>         <dbl> <int>
1 EAST          1.61     23
2 WEST          2.19     26
```

```
##running ANOVA
by.anova = aov(DISTANCE ~ DIRECTION.f, data = d.by)

summary(by.anova)
```

```
              Df Sum Sq Mean Sq F value Pr(>F)
DIRECTION.f   1    4.16   4.157    3.945 0.0528 .
Residuals    47   49.52   1.054
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##partial eta squared
by.table = tidy(by.anova)
by.table
```

```
# A tibble: 2 x 6
  term          df sumsq meansq statistic p.value
  <chr>        <dbl> <dbl>   <dbl>    <dbl>   <dbl>
1 DIRECTION.f     1  4.16   4.16     3.95  0.0528
2 Residuals     47 49.5   1.05     NA     NA
```

```

ss_residual_by = by.table %>%
  filter(term == "Residuals") %>%
  pull(sumsq)

by.table %>%
  filter(term != "Residuals") %>%
  mutate(partial_eta_sq = (sumsq)/(sumsq + ss_residual_by)) %>%
  select(term, df, statistic, p.value, partial_eta_sq)

```

```

# A tibble: 1 x 5
  term          df statistic p.value partial_eta_sq
<chr>        <dbl>    <dbl>   <dbl>         <dbl>
1 DIRECTION.f      1      3.95  0.0528         0.0774

```

Sherbourne

```

##only SHER rows
d.sher = d.tidy %>%
  filter(STN_NAME.f == "SHER")

##mean distance rating for each direction
d.sher %>%
  group_by(DIRECTION.f) %>%
  summarise(mean_rating = mean(DISTANCE),
            n = n())

```

```

# A tibble: 2 x 3
  DIRECTION.f mean_rating    n
<fct>        <dbl> <int>
1 EAST          2.77    26
2 WEST           4      25

```

```

##running ANOVA
sher.anova = aov(DISTANCE ~ DIRECTION.f, data = d.sher)

summary(sher.anova)

```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
DIRECTION.f	1	19.31	19.306	15.11	0.000305 ***
Residuals	49	62.62	1.278		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
##partial eta squared
sher.table = tidy(sher.anova)
sher.table
```

```
# A tibble: 2 x 6
  term          df sumsq meansq statistic    p.value
<chr>        <dbl> <dbl>  <dbl>    <dbl>    <dbl>
1 DIRECTION.f      1  19.3   19.3     15.1 0.000305
2 Residuals      49  62.6    1.28      NA      NA
```

```
ss_residual_sher = sher.table %>%
  filter(term == "Residuals") %>%
  pull(sumsq)

sher.table %>%
  filter(term != "Residuals") %>%
  mutate(partial_eta_sq = (sumsq)/(sumsq + ss_residual_sher)) %>%
  select(term, df, statistic, p.value, partial_eta_sq)
```

```
# A tibble: 1 x 5
  term          df statistic    p.value partial_eta_sq
<chr>        <dbl>    <dbl>    <dbl>        <dbl>
1 DIRECTION.f      1     15.1 0.000305      0.236
```

Fill in this table as you go:

Station	Paper p	Your p	Paper p^2	Your p^2
St. George	< .001	<.001	.28	.28
Spadina	.001	.001	.20	.20
Bloor-Yonge	.053	.053	.08	.08
Sherbourne	< .001	<.001	.24	.24

Step 5: Reflection

Were you able to reproduce the results you attempted to reproduce? If not, which parts could you not reproduce?

All the results reproduced with the exception of the main effect of station in the 2 x 4 ANOVA model (I got $F(3, 194) = 23.35$ instead of the reported result in the paper, $F(3, 194) = 24.10$).

How difficult was it to reproduce the results?

It didn't seem difficult to reproduce the results.

What made it hard, and what made it easy? Think about the data files, the variable names, and how much the paper told you about the analysis.

I think having a semi-clean dataset in the beginning was the most helpful. What made it easy was (i) clearly labeled column names that match pretty directly to the variable names as reported in the paper, (ii) only relevant columns included in the dataframe, and (iii) the paper stated at least the main 2 x 4 ANOVA analysis very clearly in the paper. The only part that seems a bit hard, especially for newer researchers, is that the partial eta squared calculation and meaning isn't addressed in the paper + it wasn't clearly reported that they ran one-way ANOVA models in the individual station analyses (but these are maybe things that researchers should be familiar with and not something that needs to be addressed in the paper?)

Bonus questions

1. Why is the station F different? Your F for station (23.35) is close to the paper's (24.10) but not equal, while the interaction matches exactly. Try refitting the model with the predictors in the opposite order, `DISTANCE ~ STN_NAME * DIRECTION`. Does anything change? Look back at your cell counts. What does that suggest about how the authors' software (SPSS) and R's `aov()` each divide up the variance? `docs/anova-in-r.md` explains it, but try first.

```
flipped.anova = aov(DISTANCE ~ STN_NAME.f * DIRECTION.f,
                    data = d.tidy)

summary(flipped.anova)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)						
STN_NAME.f	3	75.47	25.156	23.446	5.41e-13 ***						
DIRECTION.f	1	0.40	0.402	0.375	0.541						
STN_NAME.f:DIRECTION.f	3	52.41	17.471	16.283	1.77e-09 ***						
Residuals	194	208.15	1.073								

Signif. codes:	0	'***'	0.001	'**'	0.01	'*'	0.05	'.'	0.1	' '	1

2. Orientation or direction? The paper describes the factor as “orientation: toward, away from”. The dataset has DIRECTION: east or west. Are those the same variable? If not, which one did the authors actually put in their ANOVA? Make an `orientation` column with `if_else()` and refit the model with it to check.

```
d.orientation = d.tidy %>%
  mutate(Orientation = if_else((DIRECTION.f == "EAST" &
                                (STN_NAME.f == "B-Y" |
                                 STN_NAME.f == "SHER")),
                                "Towards",
                                if_else((DIRECTION.f == "EAST"
                                           & (STN_NAME.f == "STG"
                                                | STN_NAME.f == "SPAD")),
                                           "Away", if_else((DIRECTION.f
                                                             == "WEST" &
                                                             (STN_NAME.f == "STG" |
                                                              STN_NAME.f == "SPAD")),
                                                             "Towards", "Away"
                                                         )))) %>%
  mutate(Orientation.f = factor(Orientation))

orientation.anova = aov(DISTANCE ~ Orientation.f * STN_NAME.f,
                        data = d.orientation)

summary(orientation.anova)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Orientation.f	1	46.89	46.89	43.705	3.61e-10 ***
STN_NAME.f	3	78.38	26.13	24.351	2.03e-13 ***
Orientation.f:STN_NAME.f	3	3.01	1.00	0.934	0.425
Residuals	194	208.15	1.07		

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

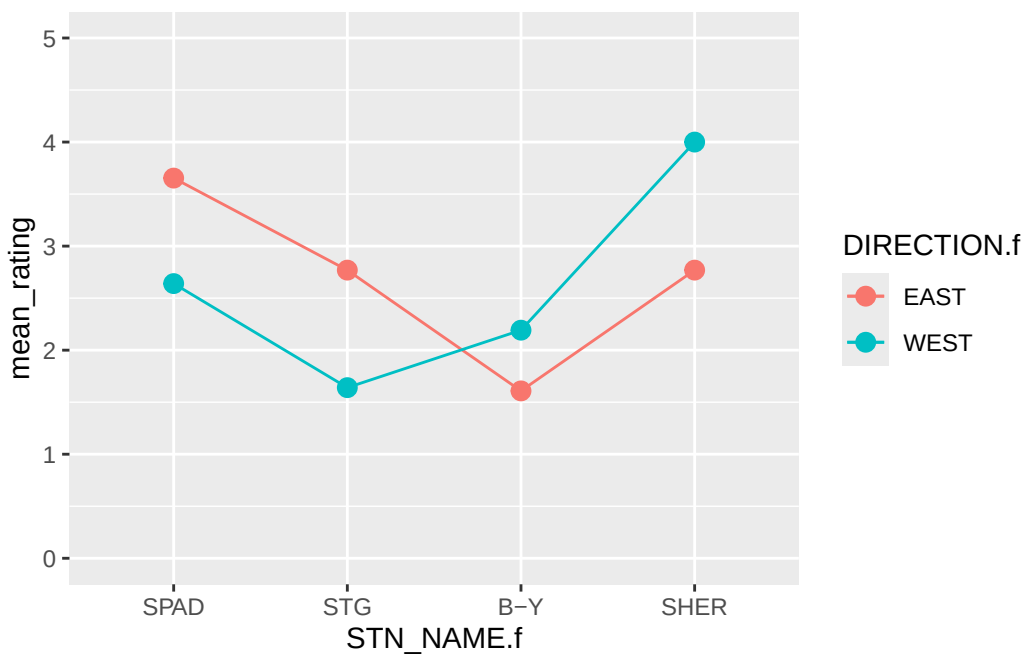
3. Reproduce Figure 1. Plot the mean distance rating for each station, split by direction, with error bars of ± 1 standard error, as in the paper. `stat_summary(fun.data = mean_se)` will draw both the means and the error bars for you.

```
d.mean = d.tidy %>%
  group_by(DIRECTION.f, STN_NAME.f) %>%
  summarise(mean_rating = mean(DISTANCE))
```

`summarise()` has regrouped the output.

- i Summaries were computed grouped by DIRECTION.f and STN_NAME.f.
- i Output is grouped by DIRECTION.f.
- i Use `summarise(.groups = "drop\_last")` to silence this message.
- i Use `summarise(.by = c(DIRECTION.f, STN\_NAME.f))` for per-operation grouping (`?dplyr::dplyr_by`) instead.

```
ggplot(d.mean, aes(x = STN_NAME.f, y = mean_rating,
                  color = DIRECTION.f,
                  group = DIRECTION.f)) +
  geom_point() +
  geom_line() +
  stat_summary(fun = "mean", geom = "point", size = 3) +
  stat_summary(fun.data = mean_se, geom = "errorbar",
              width = 0.12) +
  ylim(0,5)
```



Going further

Instead of four near-identical chunks, do all four stations in one pipeline. Look up `nest()` and `map()`.